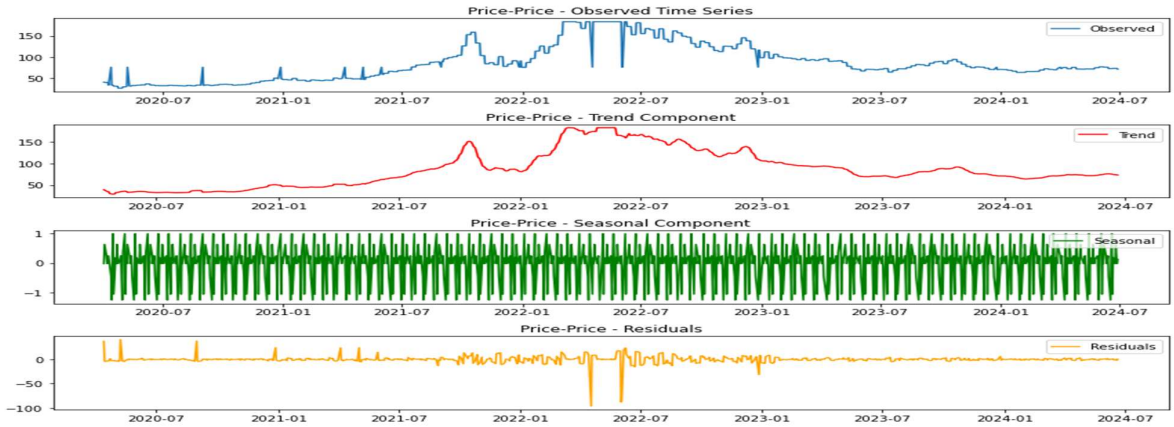
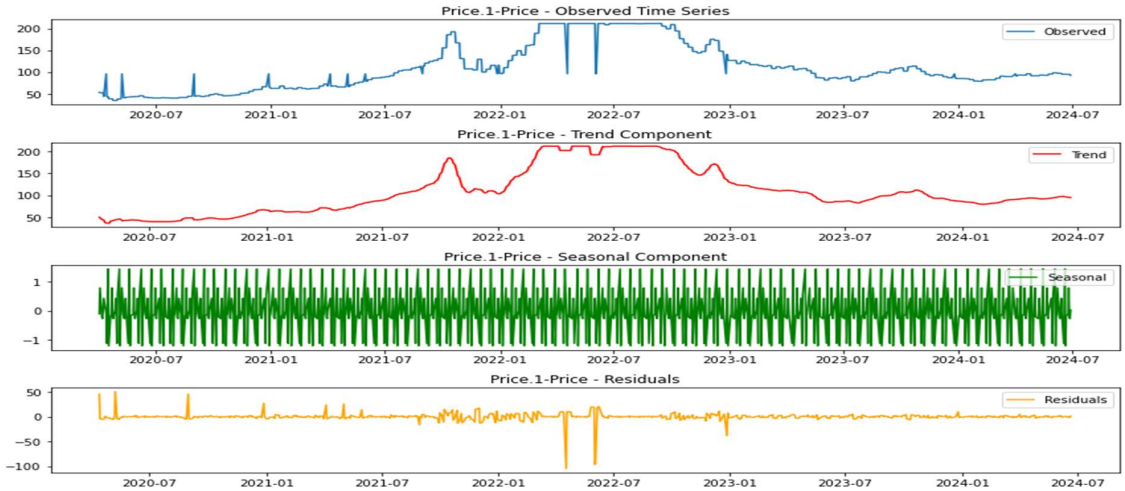


Time-Series Decomposition Plot.

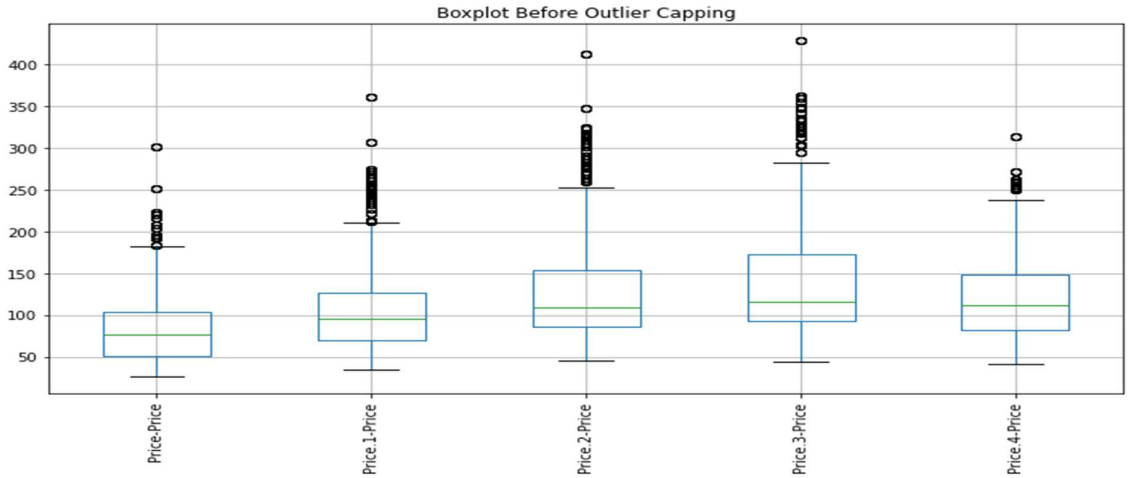
Coal prices exhibited a sharp upward trend peaking in 2022, followed by a decline, with strong seasonal patterns and occasional anomalies affecting price stability.

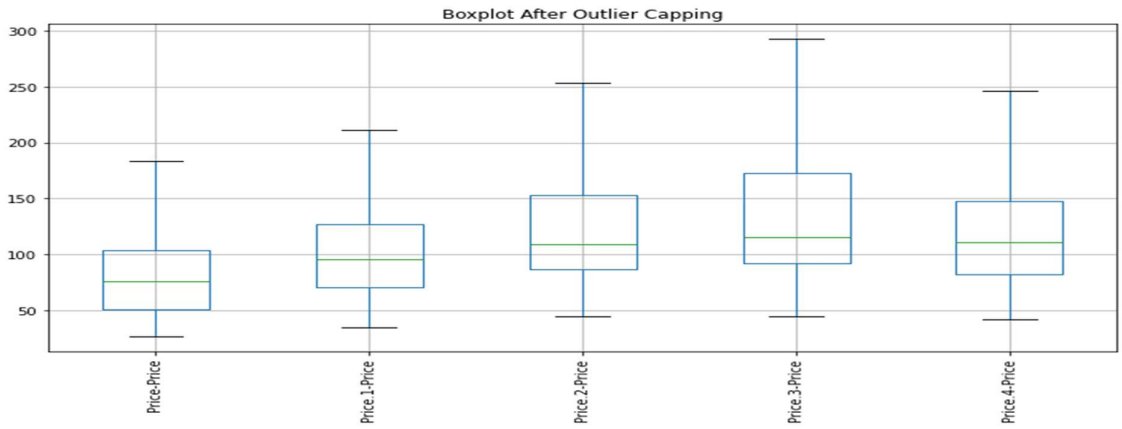
Insights from the Time Series Decomposition Plot

1. **Trend Component:** The price shows a significant rise until early 2022, followed by a decline, indicating a market boom and correction phase.
2. **Seasonal Component:** Strong periodic fluctuations suggest a recurring pattern in coal prices, likely influenced by demand cycles or external factors.
3. **Residual Component:** High residual spikes indicate anomalies or sudden external influences, particularly around early 2022.



#Box-Plot :- Univariate.





Outlier capping has effectively removed extreme values, making the data more suitable for analysis by reducing noise and potential distortions in statistical modeling.

#Scatter Plot's:-

 Outlier capping and winsorization improve data quality by reducing extreme deviations without distorting the core distribution.

 Regression models and statistical analyses will benefit from these transformations as they reduce the undue influence of outliers.

 The relationship between price variables becomes clearer, which is crucial for accurate forecasting.